5.3.7 Wake-up time from low-power modes and voltage scaling transition times

The wake-up times given in the table below are the latency between the event and the execution of the first user instruction.

The device goes in low-power mode after the WFE (wait for event) instruction

Table 31. Wake-up time from low-power modes

1. Evaluated by characterization - Not tested in production.
2. The wakeup times are measured from the wakeup event to the point in which the application code reads the first instruction.

Symbol	Parameter	Conditions	Wakeup clock	f_{HCLK} (MHz)	Typ	Max	Unit
$t_{wu(Sleep)}$	Wakeup from Sleep	Instruction cache enabled or disabled	-	-	16		CPU clock cycles
$t_{wu(Stop)}$	Wakeup from Stop 0	Flash memory in normal mode	HSI	144	3.6	5	µs
		Flash memory in low-power mode	HSI	144	7.2	11	
		Flash memory in normal mode	HSIDIV3	48	5.1	7	
		Flash memory in low-power mode	HSIDIV3	48	8.6	13.5	
	Wakeup from Stop 1(1)	Flash memory in normal mode	HSI	144	37.0	49	
		Flash memory in low-power mode	HSI	144	40.5	58	
		Flash memory in normal mode	HSIDIV3	48	39.0	53	
		Flash memory in low-power mode	HSIDIV3	48	42.0	61	
$t_{wu(Standby)}$	Wakeup from Standby mode(1)	-	HSIDIV3	48	445	-	

1. Those parameters depend on V_{CAP} capacitance value and V_{CAP} voltage at the instant of the wake-up event.

Table 32. Wake-up time using USART/LPUART

Symbol	Parameter	Condition	Typ	Max(1)	Unit
$t_{WUUSART}$/ $t_{WULPUART}$	Wake-up time needed to calculate the maximum USART/LPUART baudrate allowing to wake up from Stop mode when USART/LPUART clock source is 48 MHz by HSIDIV3	Stop 0 mode	4.8	6.6	µs
		Stop 1 mode			

1. Specified by design - Not tested in production.